



Major article

Implementation of directly observed patient hand hygiene for hospitalized patients by hand hygiene ambassadors in Hong Kong



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Key Words:

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Background: The importance of compliance with hand hygiene by patients is increasingly recognized to prevent health care–associated infections.

Methods: This descriptive study observed the effects of an education campaign, targeted to increase patients' self-initiated hand hygiene, and a hand hygiene ambassador-initiated directly observed hand hygiene program on patients' hand hygiene compliance in a university-affiliated hospital.

Results: The overall audited compliance of patients' self-initiated hand hygiene was only 37.5%, with a rate of 26.9% (112/416 episodes) before meals and medications, 27.5% (19/69 episodes) after using a urinal or bedpan, and 89.7% (87/97 episodes) after attending toilet facilities. Patients referred from a residential care home for older adults had significantly lower hand hygiene compliance ($P = .007$). Comparatively, the overall audited compliance of ambassador-initiated directly observed hand hygiene was 97.3% (428/440 episodes), which was significantly higher than patients' self-initiated hand hygiene via a patient education program (37.5%, 218/582 episodes, $P < .001$).

Conclusions: Directly observed hand hygiene can play an important role in improving compliance with hand hygiene by hospitalized patients.

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Hand hygiene is the most important component in hospital infection control. Despite continuous promotion of hand hygiene in health care workers for decades, the overall compliance remains low at 30%–40% in various clinical settings.¹ Poor hand hygiene compliance among health care workers was attributed to inadequate knowledge, heavy workload, and inaccessibility of handwashing facilities. With the introduction of alcohol-based hand rubs^{2,3} and the promulgation of the My 5 Moments for Hand Hygiene,⁴ awareness in the importance of hand hygiene in enhancing patients' safety by the prevention of health care–associated infections has heightened

among health care workers. However, the role of patients' self-initiated hand hygiene has only recently been recognized.⁵ Because the hospital environment is often contaminated with hospital-acquired pathogens, it would be important to remind and encourage patients to practice hand hygiene during their hospitalization.⁶

An education campaign on patient self-initiated hand hygiene and directly observed hand hygiene before meals and medications was promoted in our hospital beginning in 2013, as previously described.^{7,8} Because directly observed patient hand hygiene programs have been shown to reduce risks of gastrointestinal acquisition of multiple drug-resistant organisms,^{8,9} we enhanced the sustainability of our directly observed hand hygiene program through appointment of hand hygiene ambassadors in 2015. In the program, these ward-based health care assistants who were promoted as ambassadors would personally deliver alcohol-based hand rubs to all conscious hospitalized patients, with directly observed hand hygiene practice before meals and medications. The compliance with hand hygiene in the patients' self-initiated group and health care worker-assisted group was audited and analyzed.

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METHODS

This descriptive study was performed in Queen Mary Hospital, a university-affiliated hospital of 1,700 beds, where proactive infection control measures have been implemented to reduce the risk of nosocomial person-to-person transmission of epidemiologically important microorganisms.¹⁰

Self-initiated hand hygiene in patients

Patient hand hygiene practices were promoted through a continuous education campaign beginning in 2013. Each patient was given a pamphlet highlighting the importance of hand hygiene after 3 specific moments: (1) before meals and medications, (2) after using a bedpan or urinal at the bedside, and (3) after attending toilet facilities, as modified from a previous study.¹¹ Education posters were displayed by each bedside to remind patients and visitors of the importance of hand hygiene.⁷ Alcohol-based handrubs were available adjacent to each bed, toilet entrance, and toilet seat to encourage patient self-initiated hand hygiene.

To evaluate the effect of the patient education program, the compliance of self-initiated hand hygiene practice was audited by the infection control team during a 40-minute observation period per ward per day in the first 6 months of 2015 in the medical, surgical, orthopedics, and obstetrics-gynecology wards. The results of patient self-initiated hand hygiene in conscious patients were recorded according to the specified moments 1–3, as previously described, along with the epidemiologic information for further analysis. For moment 1, only self-initiated hand hygiene compliance before snacks and drinks or pro re nata medications outside meal times and regular medication rounds was included because health care worker–assisted directly observed hand hygiene of all conscious hospitalized patients before main meals and regular medication rounds was introduced in 2013.^{8,9}

Health care worker– and ambassador-assisted hand hygiene

Health care worker–assisted directly observed hand hygiene in all conscious hospitalized patients before main meals and regular medication rounds was introduced in 2013. To enhance the sustainability of directly observed hand hygiene for patients, we initiated a hand hygiene ambassador program in 2015 where ward-based health care assistants, the most frontline rank of health care workers in the clinical areas, were promoted as hand hygiene ambassadors. A 15-minute talk was given to each nominated health care assistant at the bedside to explain the significance of patient hand hygiene in infection control. To emphasize the importance of our newly appointed hand hygiene ambassadors, they were given the opportunity to meet with the infection control officer, hospital chief executive or professor of microbiology, and senior nurses

from the respective wards. A formal certificate was also presented to them in the official hand hygiene ambassador appointment ceremony. The hand hygiene ambassadors were empowered to deliver alcohol-based handrubs to all conscious patients, with directly observed hand hygiene before meals and medication rounds in their designated wards.

The compliance of ambassador-assisted hand hygiene was audited in 20 wards (6 acute medical wards, 10 surgical wards, and 4 orthopedic wards) for 22 consecutive working days in July 2015. On each audited day, 3 conscious patients were randomly selected from each of the 20 wards and interviewed to find out if they could recall ambassador-assisted directly observed hand hygiene being performed on that day. The ambassador-assisted hand hygiene was considered to have been complied if all 3 randomly selected conscious patients in the same ward reported that the ambassadors have assisted directly observed hand hygiene of these patients on that day.

STATISTICAL ANALYSIS

Fisher's exact tests was used to investigate any differences in the compliance of patient hand hygiene with respect to the 3 studied moments and age groups. The χ^2 test was used to investigate the association between hand hygiene practice and each of the epidemiologic characteristics of the patients. A *P* value of $<.05$ was considered to be statistically significant. SPSS version 23 (SPSS, Chicago, IL) was used to perform the statistical analyses.

RESULTS

Self-initiated hand hygiene in patients

Between January 14, 2015, and June 30, 2015, a total of 582 conscious patients were observed for 114 working days, with an average of 5 patients per day. The median age was 62 (range, 18–98) years, and 256 (44%) were men. Two hundred and forty patients (41%) were observed in the medical wards, 135 (23%) were observed in the surgical wards, 121 (21%) were observed in the orthopedic wards, and 86 (15%) were observed in the obstetrics and gynecology wards. Twenty-one (4%) patients were referred from residential care homes for older adults. At the time of hand hygiene audit, 301 (52%) of the patients were hospitalized for ≤ 2 days, 143 (25%) were hospitalized between 3 and 6 days, and 138 (24%) were hospitalized for ≥ 7 days.

The overall compliance of patient self-initiated hand hygiene moments 1–3 was 37.5% (218/582) according to the specified age groups (Table 1). The compliance in moment 1 (26.9%, 112/416) and moment 2 (27.5%, 19/69) was significantly lower than that of moment 3 (89.7%, 87/97; $P < .001$). With reference to the age group, patients aged ≤ 34 years tended to have lower hand hygiene compliance in moment 1 than those aged ≥ 50 years ($P = .010$), but they had higher compliance in moment 2 than those aged ≥ 80 years

Table 1
Compliance of self-initiated patient hand hygiene with respect to different age groups

Age group	Observed moment 1	Observed moment 2	Observed moment 3	Overall <i>P</i> value
≤ 34 y	12.2 (6/49)	66.7 (4/6)	90.9 (20/22)	$<.001$
35–49 y	21.0 (17/81)	33.3 (2/6)	86.4 (19/22)	$<.001$
50–64 y	34.7 (33/95)	28.6 (4/14)	81.0 (17/21)	$<.001$
65–79 y	32.7 (36/110)	30.8 (8/26)	95.7 (22/23)	$<.001$
≥ 80 y	24.7 (20/81)	5.9 (1/17)	100 (9/9)	$<.001$
Overall	26.9 (112/416)	27.5 (19/69)	89.7 (87/97)	$<.001$

NOTE. Values are percentages (compliance with hand hygiene/total observations) or as otherwise indicated.

Moment 1, hand hygiene before snacks and drinks at the bedside and before pro re nata medication; Moment 2, hand hygiene after use of a bedpan or urinal at the bedside; Moment 3, hand hygiene after attending toilet facilities.

Table 2

Epidemiologic characteristics of compliant and noncompliant patients in self-initiated hand hygiene practice during moments 1–3

Characteristic	Episodes of patients with hand hygiene practice (n = 218)	Episodes of patients without hand hygiene practice (n = 364)	P value
Age (mean ± SD), y	58.6 ± 18.0	60.6 ± 20.1	.23
Male sex	96 (44.0)	160 (44.0)	.99
Resident of elderly home	2 (0.9)	19 (5.2)	.007
Medical specialty	103 (47.2)	137 (37.6)	.023
Time of hand hygiene audit (≥7 d of hospitalization)	60 (27.5)	78 (21.4)	.10
History of hospitalization in the last 3 mo	138 (63.3)	234 (64.3)	.81
Presence of indwelling devices ^a or wounds	54 (24.8)	105 (28.8)	.29
Presence of comorbidities ^b	83 (38.1)	138 (37.9)	.97
Use of broad-spectrum antibiotics in preceding 3 mo before admission ^c	58 (26.6)	104 (28.6)	.61

NOTE. Values are n (%) or as otherwise indicated.

^aIndwelling devices include tracheostomy, drain, and urinary catheter.^bComorbidities include chronic cerebral conditions, chronic cardiac conditions, chronic pulmonary conditions, chronic renal failure, liver cirrhosis, diabetes mellitus, autoimmune diseases, and malignancies.^cBroad-spectrum antibiotics include β -lactam– β -lactamase inhibitors, cephalosporin, carbapenem, and fluoroquinolones.

($P = .010$). There was no significance difference in compliance across age groups in moment 3 ($P = .64$).

The epidemiologic characteristics of compliant and noncompliant patients with self-initiated hand hygiene in moments 1–3 were illustrated in Table 2. Patients referred from a residential care home for older adults had significantly lower self-initiated hand hygiene compliance ($P = .007$), whereas patients observed in a medical specialty had significantly higher self-initiated hand hygiene compliance ($P = .023$).

Health care worker- and ambassador-assisted hand hygiene

Nighty-six hand hygiene ambassadors were appointed in 44 wards (approximately 30 patients per ward) in Queen Mary Hospital. Hand hygiene ambassadors were responsible for the delivery of alcohol-based handrub to all conscious hospitalized patients for directly observed hand hygiene before regular meals and during medication rounds. Our audit in July 2015 revealed an overall ambassador-assisted directly observed hand hygiene compliance of 97.3% (428/440) among the selected 20 wards in Queen Mary Hospital. There were no difference in compliance in medical wards (96.2%, 127/132), surgical wards (97.7%, 215/220), or orthopedic wards (97.7%, 86/88; $P = .67$). The implementation of ambassador-assisted directly observed hand hygiene has achieved a significantly higher patient hand hygiene compliance (97.3%, 428/440) when compared with patient self-initiated hand hygiene via a patient education program (37.5%, 218/582; $P < .001$).

DISCUSSION

The overall patient self-initiated hand hygiene compliance was <40%, despite our education campaign. Our finding was similar to the historical figure of hand hygiene compliance in health care workers, which is consistently <40%.¹ Whether in patients or health care workers, hand hygiene practice is a complex human behavior that involves multiple interdependent influences from biology, the environment, education, and culture.¹² In our observations, the compliance of patient self-initiated hand hygiene was significantly higher in moment 3 than moment 1 or 2. In fact, the compliance of hand hygiene was not significantly different across age groups in moment 3 (after attending toilet facilities). It is reasonable to believe that patients, regardless of their age, perceive their hands as contaminated after toileting; therefore, self-initiated hand hygiene is mostly well complied. Young patients had significantly poorer hand hygiene compliance for moment 1 (before taking meal and medication) than those aged ≥50 years. This unexpected behavior may deserve further investigation.

The very low hand hygiene compliance among the hospitalized older adults aged ≥80 years in moment 2 (after using a bedpan or urinal at the bedside) may not represent the failure of the education campaign. These older adults may have limited mobility and therefore may not be able to access the alcohol rub without assistance. An overall lower compliance in self-initiated hand hygiene among patients from residential care homes for older adults, as shown in the epidemiologic analysis, may also be caused by the same reason. This limitation may potentially be overcome by this hand hygiene initiative, with hand hygiene ambassadors who personally deliver the alcohol-based handrub to the patient. This hand hygiene initiative could be an important intervention in the residential care homes for older adults and within the hospital.

On investigation of the epidemiologic characteristics of patients with and without self-initiated hand hygiene practice, there were no obvious clinical and epidemiologic factors that could predict compliance of self-initiated hand hygiene, except in patients staying in a medical specialty with a higher incidence of nosocomial infections. We speculated that patients with a history of health care-associated infections have higher motivation to participate in self-initiated hand hygiene, as reported in a previous study.¹³ Despite our liberal use of education posters at bedsides and in toilets to remind patients of self-initiated hand hygiene, patients' compliance in hand hygiene failed to increase with their duration of hospital stay.

With these findings and our previous success in controlling hospital outbreaks by directly observed hand hygiene, health care worker-assisted hand hygiene in patients appears to be a potential solution. A quasiexperimental study showed that the compliance of nursing staff-assisted patient hand hygiene was significantly increased from 17% to 45% after an educational intervention.⁶ In our study, we assigned ward-based health care assistants to enforce patients' hand hygiene. They were appointed as hand hygiene ambassadors and were responsible for delivering alcohol onto hands of patients and directly observing patients' handrub before meal and medication rounds. Hence, even when patients' hands are contaminated by the hospital environment, which is mutually touched by health care workers, visitors, and patients,¹⁴ patients' hands could be disinfected by alcohol before intake of meals and medications to reduce risks of gastrointestinal acquisition of multiple drug-resistant organisms. This directly observed patients' hand hygiene practice is integrated into the ambassadors' routine ward duty, and a 100% hand hygiene compliance before meals and medications can be achieved if all of the hand hygiene ambassadors play their role. This was consistent with our previous experience that directly observed hand hygiene is effective in outbreak control and prevention.^{9,15–17}

The use of hand hygiene ambassadors is an effective program to improve hand hygiene compliance among hospitalized patients through direct observation and by direct delivery of alcohol-based handrub to patients. However, it is a labor-intensive procedure. Therefore, current ambassador-assisted directly observed patient hand hygiene is only implemented before regular meals and medication rounds at day time, based on a ratio of approximately 2.2 hand hygiene ambassadors per ward of an average of 30 patients. Although we realize that other individual patient activities, especially after potentially contaminated procedures such as toileting, could also benefit from ambassador-assisted hand hygiene, there would be significant implication on manpower resources. Further observational study and administration support are required to estimate the exact number of hand hygiene ambassadors needed to cover individual episodes and to acquire necessary manpower. In addition, sustainability is a major issue in such labor-intensive procedures. Our infection control team attempted to address the issue of sustainability through regular ward visits to provide ongoing support, training, and encouragement to the ambassadors to maintain the momentum of this campaign.

There are several limitations in this study. First, patients' self-initiated hand hygiene activities are not as frequent as patient care activities by health care workers. A standardized audit protocol for patients' self-initiated hand hygiene compliance is required for subsequent investigations. Second, we conducted audits for self-initiated and ambassador-assisted hand hygiene sequentially instead of simultaneously because of the heavy workload of the infection control team. In addition, the auditing methods for self-initiated and ambassador-assisted hand hygiene were different. The change in method was to avoid bias introduced by the presence of an infection control nurse on the ward during meal time, who might act as a reminder for ambassador-assisted hand hygiene, leading to a falsely high compliance rate. Therefore, we interviewed patients to recall if they were approached by ambassadors for hand hygiene before moment 1 as a measurement of hand hygiene compliance. To minimize potential recall bias, we applied a stringent criterion to ensure compliance by asking 3 randomly selected conscious patients in the same ward to report if the ambassadors have really assisted directly observed hand hygiene of these patients on that day. Nonetheless, we realize that there is still a possibility of significant recall bias and potential deliberate bias because of reluctance of patients to report noncompliance of hand hygiene ambassadors, both of which could limit the validity of study. Therefore,

patient empowerment and an education campaign are still necessary to enhance patient hand hygiene in the health care settings.

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